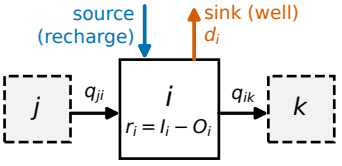
## (a) one fully mixed cell

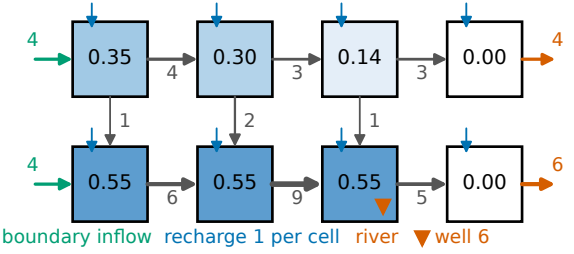



$I_i = \Sigma_j q_{ji} + \mathrm{Src}_i$

$O_i = \Sigma_k q_{ik} + \mathrm{Snk}_i$

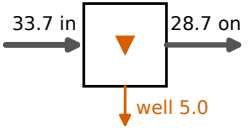
$F_i = \Sigma_k (q_{ik}/O_i) F_k + d_i/O_i$

## (b) backward fraction F to the well (gross)

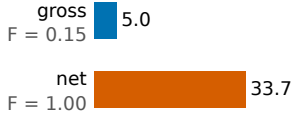



## (c) gross vs net, weak sink

weak-sink cell, tutorial 02 (m³/d)



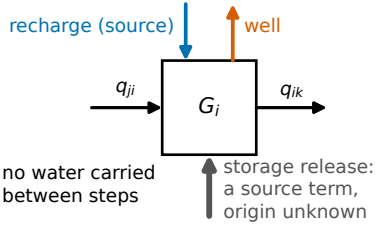

water attributed to the well (m³/d)




**net = 6.7 × pumping**

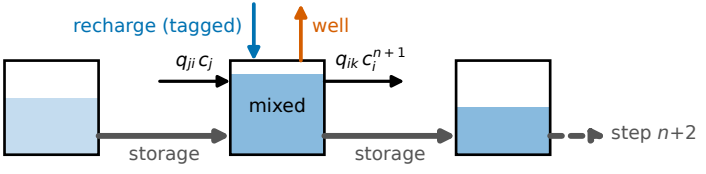
## (d) transient models: snapshot vs transient tracking

snapshot, step $n$+1




$G_i = (\Sigma_j q_{ji} G_j + s_i) / I_i$

transient tracking, step $n$ to $n$+1 (shading: tagged fraction $c$)




end of step $n$ $V_i^n,\ c_i^n$ — during step $n$+1 — end of step $n$+1 $V_i^{n+1},\ c_i^{n+1}$

$$V_i^{n+1} c_i^{n+1} = V_i^n c_i^n + \Delta t\,(\Sigma_j q_{ji} c_j^{n+1} + s_i - O_i c_i^{n+1})$$